

ESPRIT · 2026

SOFTWARE ENGINEERING

PEAXIS

A Modular AI-Powered Hiring Operating System

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Next.js

NestJS

FastAPI

PostgreSQL

pgvector

Gemini

Stripe

Docker

1 INTRODUCTION

Why PEAXIS?

Recruitment does not need another tool — it needs an integrated operating system.

01 Hiring Fragmentation

SMEs juggle 5+ disconnected tools to complete a single hire — job boards, spreadsheets, email threads, calendar apps, and manual ATS entries. There is no unified operational layer.

02 Legacy ATS Failure

Traditional ATS platforms rely on keyword matching and rigid workflows. They miss qualified candidates, penalise non-standard CVs, and provide zero intelligence — just a database with filters.

03 Candidate Friction

Candidates apply into black boxes. No feedback, no transparency, no skill gap visibility. The application experience is broken — and it silently destroys employer talent brand.

04 The Intelligence Gap

Recruiters make consequential decisions without data. No explainable scoring, no skill alignment visibility, no suggested actions. AI in recruitment today is mostly marketing — not infrastructure.

- **Core thesis:** PEAXIS unifies business infrastructure, recruiter execution, candidate experience, and AI intelligence into a single modular ecosystem — replacing the fragmented stack with an **AI-powered Hiring Operating System**.

Prospecter — Internship Host

An AI-powered B2B sales prospecting platform. I joined Prospecter full-time as **Technical Lead & Software Engineer** — leading production engineering while conducting this internship in parallel.

MISSION

Replace manual sales prospecting with a fully autonomous AI SDR — enabling B2B companies to scale outbound revenue without scaling headcount.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine

Autonomous AI sales rep — qualifies leads and orchestrates outreach sequences at scale.



Lead Intelligence

Enriches prospects with firmographics and intent signals to prioritise high-value targets.



Outbound Automation

Multi-channel sequences with LLM personalisation — zero manual prospecting.



Pipeline Analytics

Real-time conversion tracking, A/B message testing, and CRM sync.

 **Role context** — Working full-time as Technical Lead at Prospecter provided direct production exposure to LLM pipelines, multi-tenant SaaS architecture, and queue-based background processing — all of which were applied directly in designing PEAXIS.

2 INTERNSHIP CONTEXT

Building **PEAXIS** at Production Scale

Internship at Prospector — serving concurrently as **Technical Lead & Software Engineer**.

PEAXIS is an independent product conceived, architected, and built entirely as the internship project.

PEAXIS is not a Prospector product. It is a separate AI recruitment SaaS ecosystem independently designed and built during the internship period — combining academic engineering rigour with real production-grade software development practices.



Full-Stack Architecture

- NestJS API — 14 feature modules
- Next.js Core + Hire + Jobs + Landing
- Multi-tenant data model with full isolation



AI Integration

- FastAPI AI microservice
- CV parsing + match scoring engine
- Hiring Copilot · JD Generator · Bias Check



Platform Engineering

- Docker multi-stage deployment
- pgvector semantic search
- Stripe billing + Webhook integration

TECHNICAL LEAD

SOFTWARE ENGINEER

AI INTEGRATION

PLATFORM ARCHITECT

SOLO CONTRIBUTOR

3 THE PROBLEM

Recruitment is **broken** at every layer

The hiring process hasn't fundamentally changed in 20 years. Time, money, and talent are wasted at every step.

42 days

Average time-to-hire
SHRM 2024

75%

Recruiters overwhelmed by CV volume
LinkedIn Talent Report

60%

Candidates ghost after poor UX
Indeed Survey 2023



Screening bottleneck

Recruiters spend 23 hours/week manually reviewing CVs. 80% of that time is wasted on unqualified applications.



Bias & inconsistency

Manual screening introduces unconscious bias and inconsistent evaluation criteria across reviewers.



Candidate friction

Long applications, zero feedback, and opaque processes cause top talent to abandon before completion.

Global: companies lose **\$14,900** per bad hire; top candidates accept offers in under 10 days.

Regional: In Tunisia & North Africa, growing SMEs rely on spreadsheets — modern ATS adoption remains critically low.

3 THE PROBLEM

Manual workflows, legacy ATS & fragmented stacks are **not enough**

None deliver the AI-native intelligence modern hiring requires.

CAPABILITY	MANUAL	LEGACY ATS	FRAGMENTED	PEAXIS
AI-powered candidate ranking	✗	✗	partial	✓
Explainable match scores	✗	✗	✗	✓
Automatic CV parsing	✗	✗	partial	✓
Candidate-facing portal	✗	basic	basic	✓
Real-time analytics pipeline	✗	✗	✗	✓
Semantic job search	✗	✗	✗	✓
Hiring copilot / AI assistant	✗	✗	✗	✓
Multi-tenant SaaS architecture	✗	partial	✗	✓

3 THE PROBLEM

The market gap: intelligent + integrated + explainable



AI black boxes

"This candidate scored 72" — but why? Recruiters cannot act on scores they cannot explain. Bias complaints follow.



Fragmented tooling

Notion for job specs. Greenhouse for pipeline. Calendly for interviews. Zero integration — context lost at every handoff.



No candidate intelligence

Existing tools track applications. None actively guide recruiters on who to advance, how to interview, or what risks to flag.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



PEAXIS fills this exact gap

AI scores are explainable. Tools are integrated. The candidate experience is first-class. That's the OS layer missing from the market.

4 FUNCTIONAL REQUIREMENTS

Core platform requirements

FR-01 to FR-05: foundational capabilities for a production-grade SaaS recruitment system.



FR-01

Authentication & Security

- Email/password — Argon2 hashing
- OAuth2: Google + LinkedIn
- JWT access (15m) + refresh (7d)



FR-02

Multi-Tenant Architecture

- Business-scoped data isolation
- X-Business-ID header enforcement
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise
- JwtAuthGuard + PlanGuard



FR-04

Job Management

- Full CRUD with rich filters
- Semantic search via pgvector
- Auto embedding generation



FR-05

Application Pipeline

- 6 stages: Applied → Hired/Rejected
- Stage transition history
- AI scoring at submission time

4 FUNCTIONAL REQUIREMENTS

AI & analytics requirements

FR-06 to FR-10: the intelligence layer — PEAXIS's core differentiator above traditional ATS tools.



FR-06

AI Match Scoring

- Deterministic skill-overlap engine
- Score 0-100 with explanation
- Strengths & weaknesses breakdown



FR-07

CV Parsing

- PDF / DOCX / TXT support
- Gemini 2.5-flash extraction
- Auto pre-fills candidate profile



FR-08

Candidate Onboarding

- Anonymous apply-with-CV flow
- Profile completeness scoring
- Semantic embedding generation



FR-09

Interview Scheduling

- Interview CRUD with type flags
- Linked to application record
- Interviewer assignment & tracking



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking

5 NON-FUNCTIONAL REQUIREMENTS

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods behind load balancer. BullMQ handles surge via Redis queue.



OWASP aligned

NFR-02 **Security**

Argon2, JWT rotation, OAuth2, X-Business-ID isolation, OWASP-hardened API.



p95 < 200ms

NFR-03 **Performance**

Redis 3-layer caching (7d/30d/24h). Sub-200ms p95. Async BullMQ for heavy ops.



99.5% SLA target

NFR-04 **Availability**

Preferred-region SDK retries. BullMQ retry logic. Docker health checks.



Full audit trail

NFR-05 **Explainability**

Every AI score exposes: numeric value, label, matched skills, missing skills, explanation.



Microservices

NFR-06 **Modularity**

NestJS feature modules. Independent AI service. Decoupled frontend apps.

PEAXIS — Modular AI Hiring Operating System

Three deeply integrated modules replacing every tool a modern hiring team tolerates today. One ecosystem. One data layer. Every stakeholder served.



Operational Backbone

PEAXIS Core

Central business infrastructure: auth, billing, multi-tenant governance, team roles, account management.
One unified control center.



Recruiter Execution

PEAXIS Hire

Full ATS with AI assistance: pipeline Kanban, job workspace, AI match scoring, candidate summaries, interview scheduling, analytics.



Talent Acquisition

PEAXIS Jobs

Public talent platform: semantic job search, CV upload with AI parsing, match preview, skill gap visibility, easy apply, recommendations.

BUSINESS ADMIN

PEAXIS Core

Central governance — billing, team, roles, org settings

RECRUITER / HR TEAM

PEAXIS Hire

AI-assisted execution — pipelines, scoring, analytics, collaboration

CANDIDATE / TALENT

PEAXIS Jobs

Intelligent discovery — semantic search, match preview, easy apply

• All modules share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

6 SOLUTION

PEAXIS Core — Business Operations

Central business infrastructure and governance platform. Authentication, multi-tenant management, billing, and team operations — one unified control center.



Authentication & Security

JWT RS256 with Google/LinkedIn OAuth for secure access



Multi-Tenant Isolation

Complete data isolation per organization via businessId scoping



Role-Based Access

Owner, Admin, Recruiter, Viewer roles with fine-grained permissions



Billing & Subscriptions

Stripe integration with automatic subscription and webhook automation



Team Management

Invite team members, manage seats, org settings, and preferences



Account Governance

Centralized user management, activity logs, and security policies

Strategic Value: PEAXIS Core transforms fragmented administrative operations into one unified business control center for teams of any size.

JWT RS256

MULTI-TENANT

STRIPE BILLING

ROLE-BASED ACCESS

3 SOLUTION

PEAXIS Hire — Recruiter Execution

Complete ATS with AI-assisted workflows. Centralizes recruiter operations into one seamless platform from job creation to offer.



Pipeline Kanban

Drag-drop pipeline management with AI match scores on every candidate card



Hiring Copilot

AI summaries, interview questions, risk flags, skill gaps, suggested actions



Job Analytics

Funnel metrics, conversion rates, time-to-stage, score distributions per job



Job Workspace

Central hub — job details, pipeline, candidates, analytics, team collaboration

4 SOLUTION

PEAXIS Jobs — Talent Acquisition

Public talent acquisition platform designed to optimize candidate discovery, onboarding, and application experience.



Semantic Job Search

pgvector embeddings
find relevant jobs
beyond keyword
matching



Job Lite Format

Lightweight public job
listings optimised for
mobile discovery



AI CV Parsing

Upload CV → Gemini
parses → candidate
profile auto-filled
instantly



Match Score Preview

See AI match
percentage + skill gaps
before applying —
transparency first

7 TECHNOLOGIES

Tech Stack — Four Layers

Production-grade architecture. Modular, scalable, and designed for real-world hiring at scale.

Frontend

PEAXIS Core (port 3000) · Hire (3101) · Jobs (3100) — Server components with interactive client state

Next.js 16

React

Tailwind

TypeScript

API

Core API (port 4000) · 14 modules · 60+ endpoints — Single source of truth for all data

NestJS

Prisma

JWT

TypeScript

AI Service

Queue-driven via BullMQ · CV parsing, match scoring, JD generation — Cacheable in Redis

FastAPI

Gemini

Python

pgvector

Database

PostgreSQL + 1536-dim embeddings · Redis multi-layer cache (7d/30d/24h) · Async job queue

PostgreSQL

pgvector

Redis

BullMQ

MULTI-TENANT

Every query scoped to businessId

ASYNC PROCESSING

BullMQ + background workers

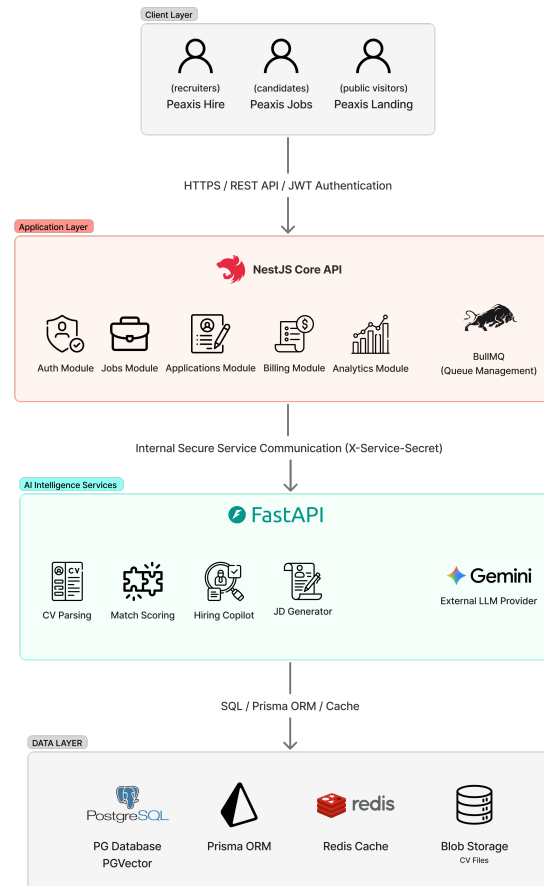
SEMANTIC SEARCH

pgvector + 1536-dim embeddings

8 ARCHITECTURE

Logical Architecture

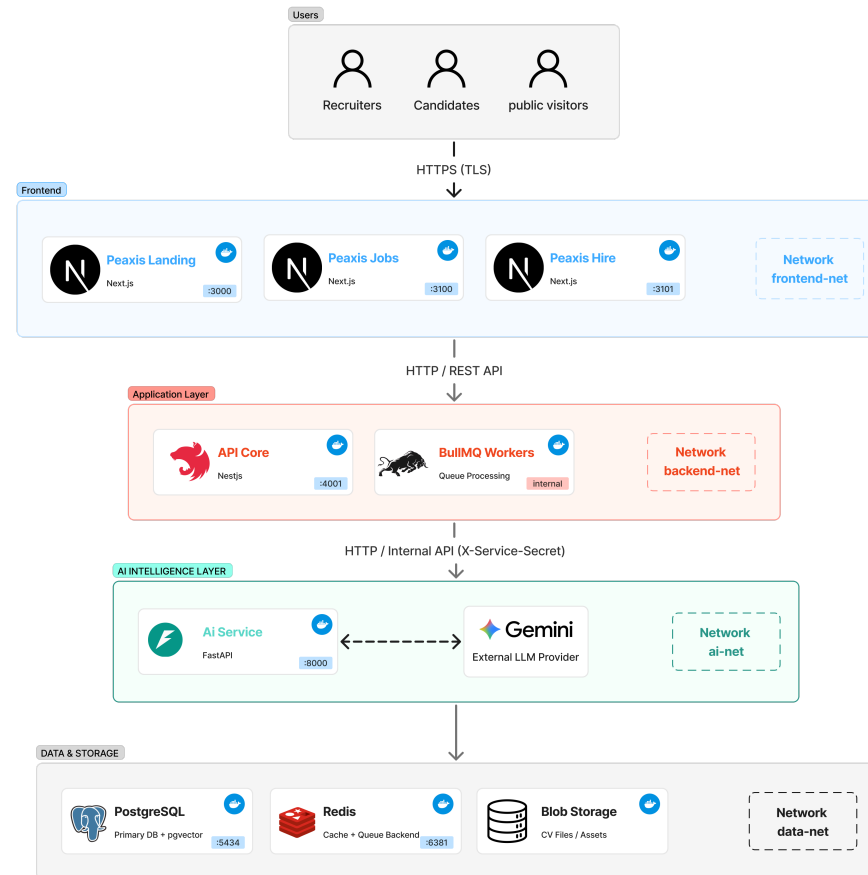
4-layer design · Client → Application → AI Intelligence → Data



9 ARCHITECTURE

Physical Architecture

Docker Compose deployment · containerised services · network segmentation



PEAXIS: Achievement & Future

14

API Modules

60+

REST Endpoints

6

AI Features

4

Microservices



Key Achievements

- End-to-end multi-tenant SaaS with three consumer apps + business platform
- Explainable AI scoring engine with matched/missing skill breakdown
- Semantic search powered by pgvector embeddings
- Queue-based architecture supporting async background processing
- Production-ready Stripe billing integration with webhook automation
- Full API documentation via Swagger — 60+ endpoints across 14 modules



Technical Lessons

- Multi-tenant isolation must be enforced at the guard level — retrofitting is 10x harder
- Separate "what" (deterministic) from "why" (LLM) for cost-effective AI at scale
- Server components + router.refresh() unlock seamless workspace UX without full navigations
- Hybrid scoring (skill-overlap + LLM explanation) beats pure LLM on cost and reproducibility
- Cache-first with intelligent TTLs (7d / 30d / 24h) reduces AI costs by 80%



Future Roadmap

- Real-time collaborative hiring workspace with role-based decision tracking
- Multilingual CV parsing and region-aware job matching (MENA market focus)
- Advanced analytics dashboard with predictive pipeline velocity metrics
- Integration ecosystem — LinkedIn Talent Solution, Greenhouse, Workday connectors
- Mobile candidate app for on-the-go application management



PEAXIS demonstrates that AI-powered recruitment can be built with engineering rigour, academic depth, and real product maturity — transforming a 6-month internship into a fully deployable, multi-tenant SaaS platform ready for market.

Thank you For your time

Contact: BIBANI Mohamed Habib Allah — contact@peaxis.com · Questions welcome
after the session.

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